

Dr. Mousumi Ahmed Mimi

Dynamic and accomplished Assistant Professor in Computer Science and Engineering with a PhD in Data Science from Multimedia University, Malaysia (completed in 2.8 years). I specialize in machine learning and deep learning applications for network security and threat detection, with a focus on near real-time Android spyware detection and prediction using network traffic analysis. My research has resulted in multiple peer-reviewed publications and the development of practical AI-driven security solutions. I am currently seeking a physical postdoctoral research position in Germany in the areas of Data Science, Machine Learning, Deep Learning, or AI for any topic. With proven quick learning ability, strong analytical skills, and a track record of delivering high-impact research alongside effective teaching and student mentoring, I aim to contribute to cutting-edge projects while building international collaborations. This experience will further strengthen my expertise and position me for a future faculty role in a leading academic institution. A fast learner and adaptable researcher, I am passionate about bridging innovative AI research with real-world applications and quality teaching.

CONTACT

Address: 199/4, South Paik Para, Mirpur, Dhaka Bangladesh
Cell: +8801540086347
Mail: mousumimimi609@gmail.com
mousumi_cse@iubat.edu

SKILLS

- Data Science
- AI
- Machine Learning
- Deep Learning
- VANET
- Cyber Security
- Software Development (Python, .net core, Java, php)
- Java Script, HTML, CSS
- Research
- Giving lecture to my students in English
- Giving Lecture and communicate with foreign students.

LANGUAGES

- **English**: Fluent in speaking , writing, listening and reading
- **Bangla**: Fluent in speaking , writing, listening and reading .
- **Hindi**: Fluent in speaking, listening

EXPERIENCE

- **2026 January – Continue-** Assistant Professor in Computer Science and Engineering at IUBAT-International University of Business Agriculture and Technology.
- **2023 February-2025 November-** PhD in Data Science Research, specializing in Machine Learning and Deep Learning.
- **2020 January-2022 December-** Lecturer in Computer Science and Engineering at IUBAT-International University of Business Agriculture and Technology.
- **August-2018- December-2018-** Software Development intern at Virgo Global (Developed Inventory Management System for Virgo Global)

PhD Research Highlights

Title: Near real-time detection of android spyware leveraging directional network traffic features and rule-based labelling

Key contributions:

- Collected and analyzed network traffic packets from 14 distinct Android spyware types, along with normal traffic for baseline comparison.
- Performed packet-level analysis using Wireshark to extract relevant behavioral patterns.
- Developed datasets based on directional network traffic feature engineering and rule-based labelling for classification.
- Applied and evaluated multiple machine learning and deep learning models, including:
  - XGBoost, Random Forest, CatBoost, LightGBM
  - CNN, LSTM, FFNN, TabNet
  - Hybrid models such as FFNN-XGB, TabNet-XGB, CNN-LSTM, and LSTM-XGB
- Conducted real-time detection and prediction experiments using multiclass classification techniques.
- Designed and implemented a prototype user interface for practical deployment and demonstration.

Master's Research Highlights

- Ethical Hacking
- Digital Forensics
- Network Security
- Advanced Computer Communications and Networks
- Analysis of malicious network traffic using WireShark
- Simulation using SUMO and NS3

- Research on VANET

Teaching Contributions as Lecturer

The following courses I took:

- C, C++, C#, JAVA, .net core
- Software Engineering
- System Analysis and Design
- Management Information System
- Data structure and Algorithms
- Computer Fundamentals
- Supervised thesis students (IoT, Machine learning, deep learning, VANET)
- Supervised practicum students (software development)

- Attending conferences to learn about new research and present findings.

Scholarships, Awards and Professional Certifications

- CNE-6 Gold (Doing free physical classes) –2025
- CSE-6 Gold (Doing free physical classes) – 2025
- IPv6 Security Workshop 2025
- 2nd Global IPV6 Cyber Security Seminar-2025
- MMU Archiver Research Scholarship
- Sustainable Computing and Management SUSCOM-2022
- 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)
- Gold Medal for Academic Excellence Award– 2018
- IUBAT Academic Excellence Award – 2017
- IUBAT Academic Excellence Award – 2016
- IUBAT Academic Excellence Award – 2015

EDUCATION

2023-2025	PhD in Data Science Title: Near real-time detection of android spyware leveraging directional network traffic features and rule-based labelling	Pass with Minor Correction
2021-2019	M. Sc in Computer Science and Engineering in UIU (United International University)	CGPA 3.89
2018-2015	BCSE in IUBAT (International University of Business Agriculture and Technology)	CGPA- 4.00

PUBLICATIONS

- Mimi, M. A., Ng, H., & Yap, T. T. V. (2025). Directional features and rule- based labeling for real-time network traffic-basedandroid spyware classification: MA Mimi et al. The Journal of Supercomputing, 81(8), 932.

- Mimi, M. A., Ng, H., Yap, V., & Tzen,T. (2025). Curating datasets to enhance spyware classification. Jordanian Journal of Computers & Information Technology, 11(1).
- M. A. Mimi, T. T. V. Yap, and H. Ng, "Advancements in Machine Learning Techniques for Intrusion Detection Systems: An Overview of Perspectives and Datasets," Journal of Logistics, Informatics and Service Science, vol. 11, no. 3, 2024.
- Mimi, M. A. (2022). Using clustering scheme: Prevent reply attack in vehicular ad-hoc networks (VANET). Journal of Discrete Mathematical Sciences and Cryptography, 25(7), 2007-2014
- Mimi, M. A. (2022). A Stable Clustering Architecture for Efficient Routing and Resource Allocation in Vehicular Ad Hoc Networks (Doctoral dissertation, UIU).
- Mimi, M. A., & Elahi, M. M. (2021, July). A Stable Clustering Architecture for Efficient Routing in Vehicular Ad Hoc Networks. In 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI) (pp. 1-6). IEEE.

Papers Under Review

- MAS-2024: Collection and Development of Large-Scale Android Spyware Datasets from Network Traffic for Multi-Class Classification
- Mutual Information-Based Learning Models for Differentiating Alzheimer's Disease Stages Using EEG Signals.
- Adaptive Learning in Intrusion Detection Systems: A Review (Accepted)
- A Systematic Review of Explainable AI Techniques for Android Malware and Spyware Detection
- SHAP-Driven Liver Cirrhosis Prediction with Balanced Learning
- Key Contributions of Explainable Ensemble Learning for Breast Cancer Detection
- Comparative Oversampling and Explainable Machine Learning for Type-2 Diabetes Prediction
- Explainable Parkinson's Disease Severity Prediction Using Consensus Feature Voting